

Raonament Basat en Casos.

Aplicacions Acadèmiques i Industrials

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Aplicacions acadèmiques de CBR

- **Diagnosis**

- PROTOS [Porter & Bareiss, 1986]
- CASEY [Coton, 1989]
- CASCADE [Simoudis, 1992]
- PAKAR [Watson & Abdullah, 1994]

- **Planificació**

- BATTLE [Goodman, 1989]
- BOLERO [López & Plaza, 1993]
- TOTLEC [Costas & Kayshap, 1993]

- **Raonament Legal**

- ◆ JUDGE [Bain, 1986]
- ◆ HYPO [Ashley, 1988]
- ◆ KICS [Yang & Robertson, 1994]

- **Disseny**

- ◆ CYCLOPS [Navinchandra, 1988]
- ◆ JULIA [Hinrichs, 1989]
- ◆ CADET [Sycara, 1992]
- ◆ ARCHIE [Pearce *et al.*, 1992]
- ◆ NIRMANI [Perera & Watson, 1995]

■ Raonament per Analogia

- ◆ SWALE [Kass, 1989]

■ Arbitratge

- ◆ MEDIATOR [Simpson, 1985]
- ◆ PERSUADER [Sycara, 1987]
- ◆ KICS [Yang & Robertson, 1994]

■ Adaptació

- ◆ CHEF [Hammond, 1986]
- ◆ PLEXUS [Alterman, 1986]
- ◆ COACH [Collins, 1987]

■ Sistemes Tutors

- ◆ DECIDER [Farrel, 1987]
- ◆ HYPO [Alevén & Ashley, 1992]

■ Supervisió

- ◆ CASEPLANT [Sánchez-Marrè, 1996]

PROTOS

[Porter & Bareiss, 1986]

- *Task*: heuristic classification
- *Domain*: audiology
- *Automated/Interactive*: automated reasoning with interactive knowledge acquisition in the event of failures
- *Indexing*: learned associations between situational features and classifications, learned prototypicality ratings of cases, learned differences between similar cases
- *Retrieval algorithm*: classification algorithm
- *Organisation of cases*: organized in categories and ranked in order of prototypicality; difference links connect similar cases with significant differences
- *Matching process*: knowledge-based pattern matching

CASEY

[Koton, 1989]

- *Task*: explanation, diagnosis
- *Domain*: heart failures
- *Automated/Interactive*: automated
- *Indexing*: importance of features to the causal explanation of each patient
- *Retrieval algorithm*: redundant discrimination network
- *Organisation of cases*: prioritised redundant discrimination network
- *Matching process*: evidence heuristics and a causal model
- *Case selection*: causal model
- *Adaptation methods*: model-guided repair

BATTLE

[Goodman, 1989]

- *Task*: projecting the results of actions for plan evaluation
- *Domain*: land warfare battle planning
- *Automated/Interactive*: interactive
- *Indexing*: inductively derived indexes based on simple qualitative model of the domain
- *Retrieval algorithm*: binary decision-tree traversal
- *Organisation of cases*: two-tiered memory organisation: expert-defined categories and binary decision-trees
- *Case selection*: user decision

HYPPO

[Ashley, 1988]

- *Task*: case-based interpretation and argumentation
- *Domain*: legal reasoning, trade secrets law
- *Automated/Interactive*: mostly automated
- *Indexing*: dimensions (predictive factors for a domain)
- *Retrieval algorithm*: retrieve any case indexed by a dimension that applies to or is near miss to the problem
- *Organisation of cases*: flat, indexed by dimensions
- *Case selection*: sort cases into a claim lattice, and select cases that share maximal subsets of dimensions with problem

JULIA

[Hinrichs, 1989]

- *Task*: design
- *Domain*: meal planning
- *Automated/Interactive*: automated, with user selecting preferences
- *Indexing*: specifications in problem descriptions
- *Retrieval algorithm*: searching in redundant discrimination networks
- *Organisation of cases*: redundant discrimination networks
- *Matching process*: heuristic matching rules
- *Case selection*: preference procedure
- *Adaptation methods*: common-sense transformation, special-purpose structure-modifying heuristics

ARCHIE

[Pearce *et al.*, 1992]

- *Task*: aiding design
- *Domain*: architectural design (office buildings)
- *Automated/Interactive*: interactive
- *Indexing*: design goals, outcomes, and situation descriptors
- *Retrieval algorithm*: nearest neighbour and qualitative retrieval
- *Organisation of cases*: cases clustered around similar design goals and outcomes
- *Matching process*: nearest-neighbour
- *Case selection*: highest match score, but user can browse through all retrieved cases
- *Adaptation methods*: none

SWALE

[Kass, 1989]

- *Task*: explanation of anomalous events by analogy
- *Domain*: death and destruction
- *Automated/Interactive*: automated
- *Indexing*: anomaly characterisations, unusual features
- *Retrieval algorithm*: retrieval based on indexes
- *Organisation of cases*: flat memory one-level deep indexed
- *Case selection*: relevance to anomaly, plausibility of suggested explanation, usefulness
- *Adaptation methods*: local search, specialised search, search heuristics indexed by anomaly vocabulary

PERSUADER

[Sycara, 1987]

- *Task*: generating compromise solutions
- *Domain*: labour negotiations
- *Automated/Interactive*: automated, with feedback from users
- *Indexing*: predictive features in labour negotiation
- *Retrieval algorithm*: searching in redundant discrimination networks
- *Organisation of cases*: redundant discrimination networks
- *Case selection*: distance in multidimensional similarity hierarchies
- *Adaptation methods*: parameter adjustment, case-based adaptation

CHEF

[Hammond, 1986]

- *Task*: planning
- *Domain*: recipe creation
- *Automated/Interactive*: automated
- *Indexing*: prioritised constraints and factors known to cause failures
- *Retrieval algorithm*: discrimination net search
- *Organisation of cases*: several prioritised discrimination nets, one for each reasoning task
- *Matching process*: exact match while traversing discrimination net
- *Case selection*: none
- *Adaptation methods*: reinstantiation, special-purpose critics, special-purpose repair heuristics

CASEPLANT

[Sánchez-Marrè, 1996]

- *Task*: supervision (diagnosis and actuation) over a real-time process
- *Domain*: wastewater treatment plants
- *Automated/Interactive*: both
- *Indexing*: predictive list of features, meta-cases
- *Retrieval algorithm*: partial-matching traversal of the prioritised discrimination tree(s)
- *Organisation of cases*: prioritised discrimination tree(s)
- *Matching process*: qualitative-quantitative weighted distance measure (*L'Exemple* distance)
- *Case selection*: less distant, more similar among retrieved cases and the current one
- *Adaptation methods*: parameter adjustment

Aplicacions industrials de CBR

- **CLAVIER** [Henessy & Hinkle, 1992]
- **CASEline** [Magaldi, 1994]
- **Top Management Fraud** [Curet & Jackson, 1995]
- **The GizmoTapper** [Broderbund Software, 1995]

CLAVIER (1)

[Henessy & Hinkle, 1992]

- *Task*: design and evaluation
- *Domain*: autoclave loading, spatial arrangements
- *Industry*: Lockheed Missiles and Space Company. Sunnyvale, California
- *Automated/Interactive*: interactive
- *Indexing*: parts, part locations, and part orientations
- *Retrieval algorithm*: serial search
- *Organisation of cases*: flat memory
- *Case selection*: based on number of parts matched and priorities of matched parts, user-based decision
- *Adaptation methods*: operator graphical edition
- *Implementation*: Common Lisp, 1st version in September 1990, 20 initial cases to several hundred
- *Impact*: Transferring expertise in the company, improving knowledge acquisition, remembering is useful, 90% competence, huge economical saving

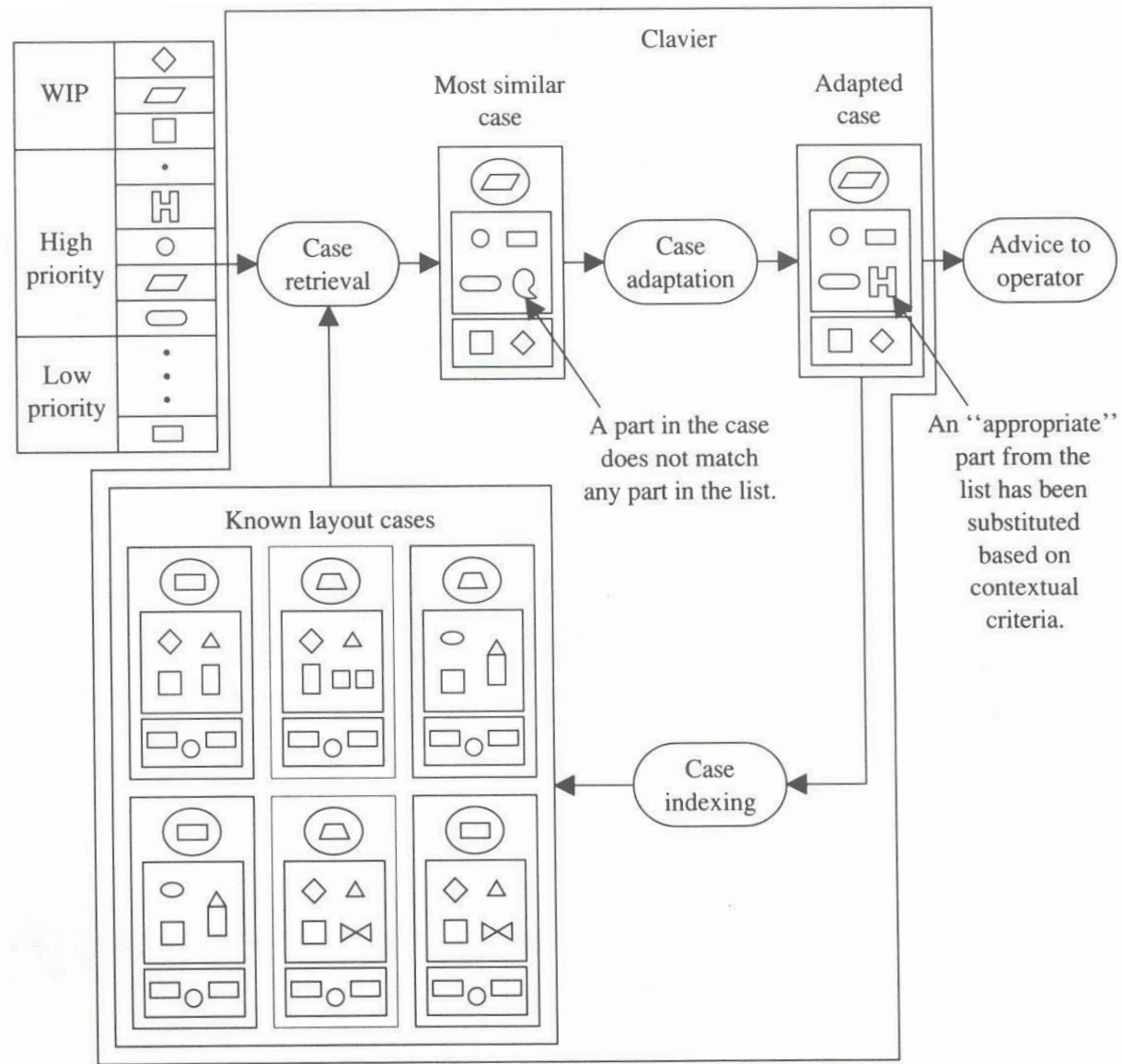


FIGURE 2.6 A Schematic of CLAVIER's Reasoning Process. Adapted from Figure 3 in Hennessy, D.H. and Hinckle, D. (1992). "Applying Case-Based Reasoning to Autoclave Loading," *IEEE Expert*, 7(5), pp. 21-26.

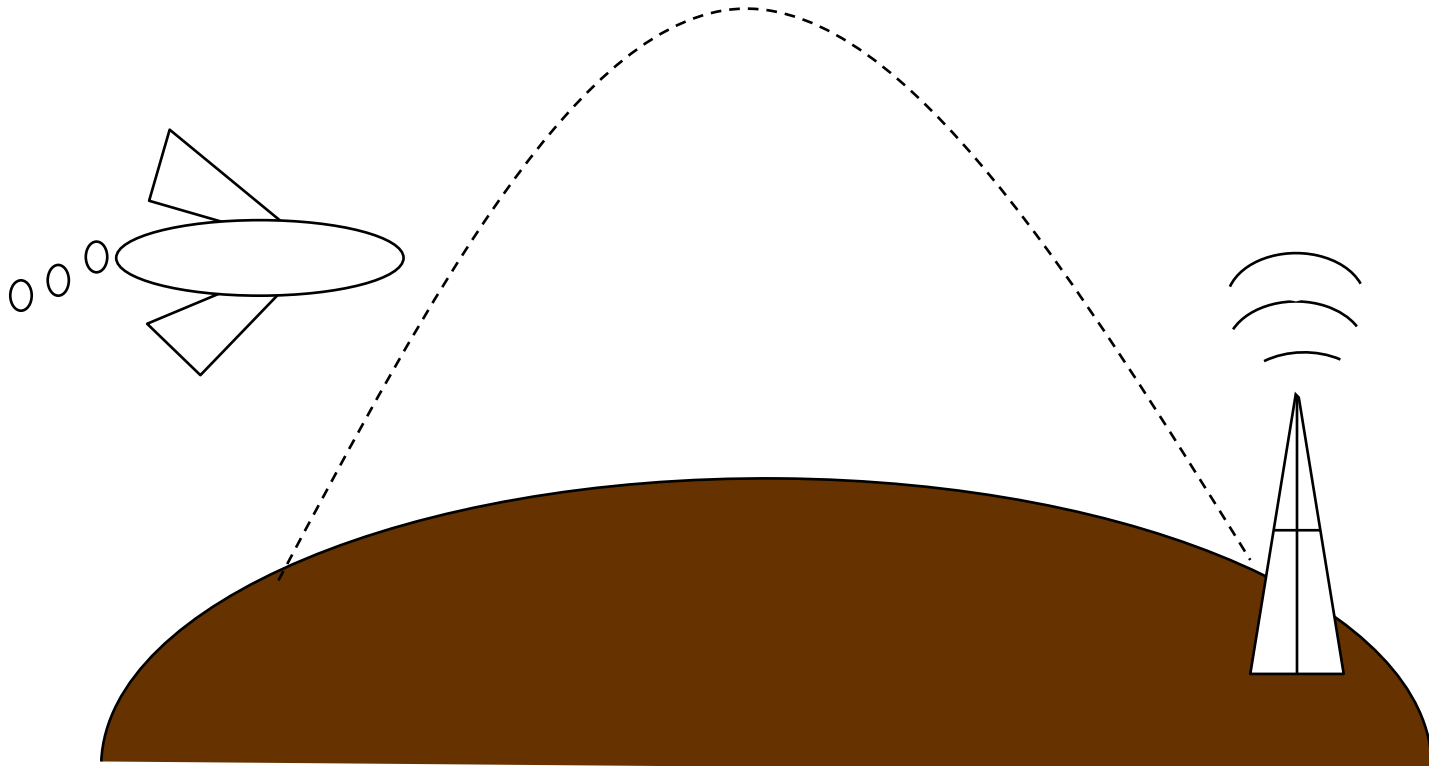
CASELine (1)

[Magaldi, 1994]

- *Task*: fault diagnosis and repair
- *Domain*: Boeing 747-400 aircraft fault diagnosis and repair between aircraft arrival and departure
- *Industry*: British Airways
- *Automated/Interactive*: interactive
- *Retrieval algorithm*: nearest-neighbour, inductive retrieval
- *Implementation*: Using Cognitive Systems' Remind in 1994. Initial 200 cases
- *Impact*: Reducing costly delays at airports, complementing human reasoning, legal defence in negligence affairs, extending to Concorde (expertise preservation)

CASELine (2)

[Magaldi, 1994]



Each minute an airplane is stopped they lose 3000 \$!!!

TMFDT

[Curet & Jackson, 1995]

- *Task*: diagnosis, likelihood assessing
- *Domain*: Top Management Fraud Discovering within a company
- *Industry*: Deloitte & Touche (multinational auditors firm of accountants)
- *Automated/Interactive*: interactive
- *Retrieval algorithm*: nearest-neighbour, inductive indexing
- *Organisation of cases*: 160 case features, knowledge acquisition from consensued questionnaires
- *Implementation*: using Cognitive System's Remind. Several hundred cases
- *Impact*: flexibility, consistent decision support, justification, expert knowledge transferred, learning capabilities, reinforcing firm's public image and competitiveness

The GizmoTapper

[Broderbund Software, 1995]



- **Task:** fault diagnosis and repair
- **Domain:** help desk customer service for games. 1st game was *Myst*
- **Industry:** Broderbund Software (a major computer games and educational software supplier). Broderbund was purchased by The Learning Company in 1998.
- **Automated/Interactive:** interactive Web-based user's interface
- **Case selection:** certainty-based selection of ranked cases
- **Implementation:** Uses Inference Corp.'s CBR3, 1st version in December 1995, 3000 initial cases.
- **Impact:** Enabling more customer support problems than human staff could, huge economical saving (equivalent to annual 7 full-time support representatives), Web-based support, customers liked the system, 94% average certainty, *Highly Commended award at the Voice Europe 1996 Exhibition in London.*

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